Surface-Mount Glass Passivated Rectifier



SMC (DO-214AB)

Cathode ○—|◀—○ Anode

LINKS TO ADDITIONAL RESOURCES

3D Models

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	15 A
V_{RRM}	400 V, 600 V, 800 V, 1000 V
I_{FSM}	450 A
I_R	5 µA
V_F at I_F = 15 A (T_J = 125 °C)	0.88 V
T_J max.	150 °C
Package	SMC (DO-214AB)
Circuit configurations	Single

FEATURES

- Low profile package
- Ideal for automated placement
- Glass passivated pellet chip junction
- Low forward voltage drop
- Low leakage current
- High forward surge capability
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available
  - Automotive ordering code: base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

TYPICAL APPLICATIONS

For use in general purpose rectification of power supplies, inverters, converters, and freewheeling diodes for consumer, automotive, and telecommunication.

MECHANICAL DATA

Case: SMC (DO-214AB)

Molding compound meets UL 94 V-0 flammability rating Base P/N-M3 - halogen-free, RoHS-compliant

Base P/NHM3 - halogen-free, RoHS-compliant, and AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102

M3 and HM3 suffix meets JESD 201 class 2 whisker test

Polarity: color band denotes cathode end

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER	SYMBOL	S15CLG	S15CLJ	S15CLK	S15CLM	UNIT
Device marking code		15LG	15LJ	15LK	15LM	
Maximum repetitive peak reverse voltage	V_{RRM}	400	600	800	1000	V
Maximum RMS voltage	V_{RMS}	280	420	560	700	V
Maximum DC blocking voltage	V_{DC}	400	600	800	1000	V
Maximum average forward rectified current	$I_{F(AV)}$ (1)	15				A
	$I_{F(AV)}$ (2)	2.1				A
Non-repetitive peak forward surge current 8.3 ms single sine-wave on rated load	I_{FSM}	450				A
Non-repetitive peak forward surge current 1.0 ms single sine-wave, T_J = 25 °C		900				
Operating junction and storage temperature range	T_J, T_{STG}	-55 to +150				°C

Notes

(1) Mounted on aluminum PCB 3 cm x 3 cm with infinite heatsink

(2) Free air, mounted on recommended copper pad area

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Instantaneous forward voltage	I_F = 7.5 A	T_J = 25 °C	V_F (1)	0.91	-	V
	I_F = 15 A			0.98	1.03	
	I_F = 7.5 A	T_J = 125 °C		0.79	-	
	I_F = 15 A			0.88	-	
Reverse current	Rated V_R	T_J = 25 °C	I_R (2)	-	5	µA
		T_J = 125 °C		96	-	
Typical reverse recovery time	I_F = 0.5 A, I_R = 1.0 A, I_{rr} = 0.25 A		t_{rr}	5	-	µs
Typical junction capacitance	4.0 V, 1 MHz		C_J	126	-	pF

Notes

(1) Pulse test: 300 µs pulse width; 1 % duty cycle

(2) Pulse test: pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**SYMBOL**	**S15CLG**	**S15CLJ**	**S15CLK**	**S15CLM**	**UNIT**
Typical thermal resistance	$R_{\theta JA}$ (1)(2)	87				°C/W
	$R_{\theta JM}$ (3)	3				

Notes

(1) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$

(2) Thermal resistance junction-to-ambient to follow JEDEC® 51-2A, device mounted on FR4 PCB, 2 oz., standard footprint

(3) Thermal resistance junction-to-mount to follow JEDEC® 51-14 transient dual interface test method (TDIM)

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PREFERRED PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
S15CLJ-M3/I	0.257	I	3500	13" diameter plastic tape and reel
S15CLJHM3/I (1)	0.257	I	3500	13" diameter plastic tape and reel

Note

(1) AEC-Q101 qualified

RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

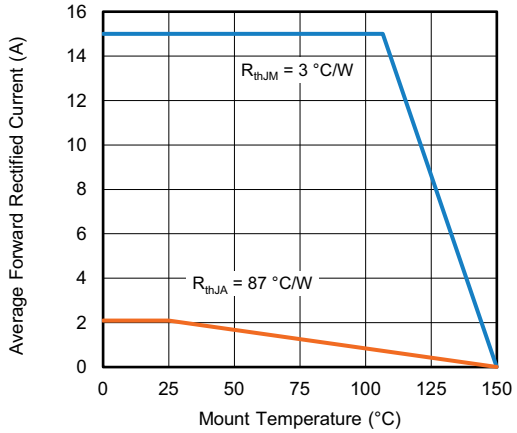


Fig. 1 - Forward Current Derating Curve

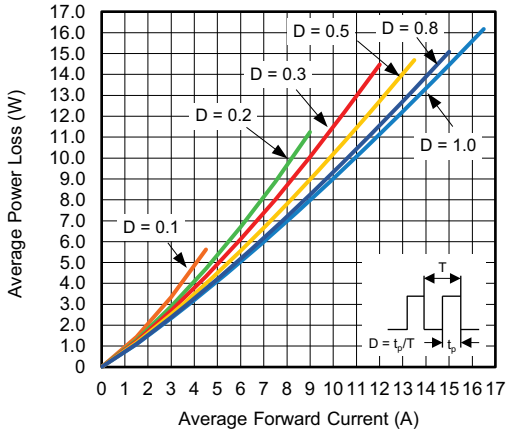


Fig. 2 - Average Power Loss Characteristics

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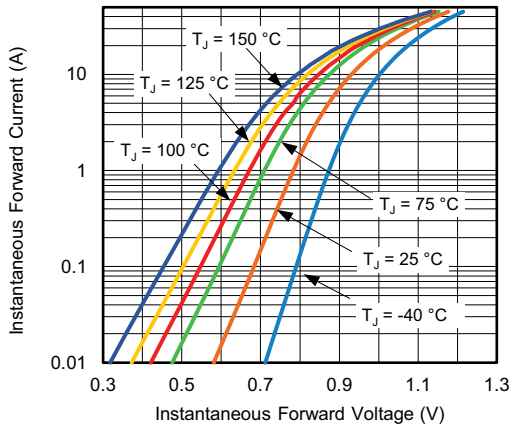


Fig. 3 - Typical Instantaneous Forward Characteristics

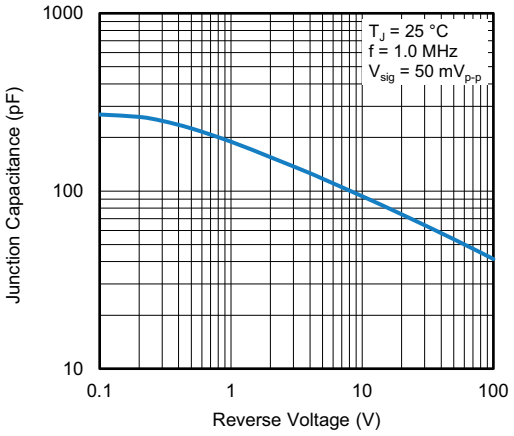


Fig. 5 - Typical Junction Capacitance

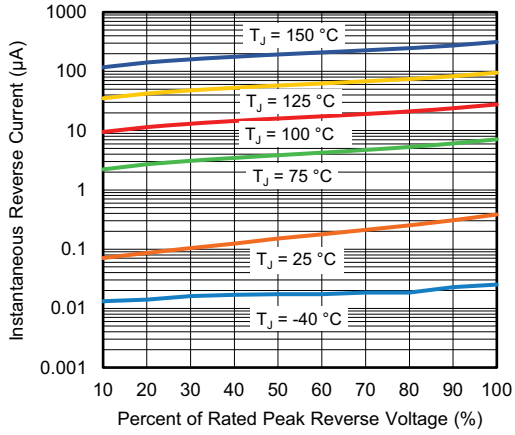


Fig. 4 - Typical Reverse Characteristics

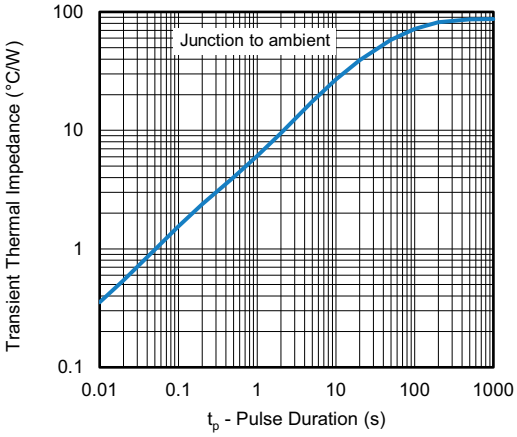
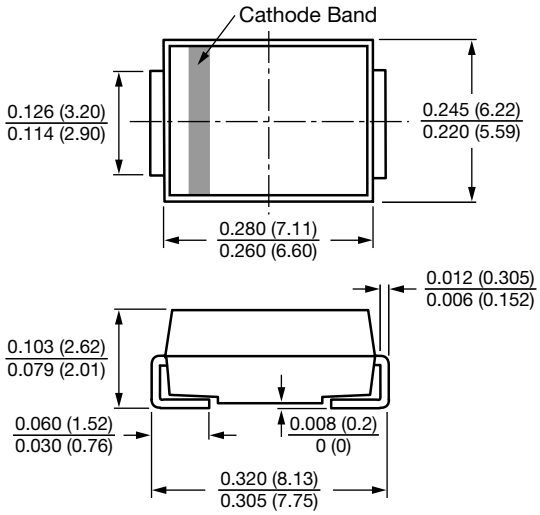


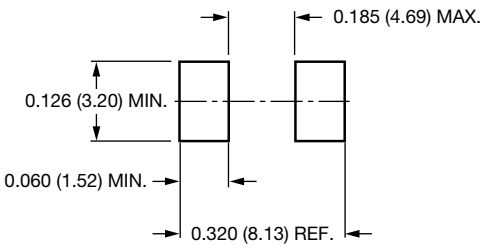
Fig. 6 - Typical Transient Thermal Impedance

PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

SMC (DO-214AB)



Mounting Pad Layout



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